

THE UNIVERSITY OF AZAD JAMMU & KASHMIR
MUZAFFARABAD

DATABASE MANAGEMENT SYSTEM LAB

Lab 11 — Scalar Functions — Numeric & Date/Time
Customer/Product & Employee Databases

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Roll No	2024-SE-11
Semester	04
Course Title	Database Management System (CS-2204)
Submitted To	Engr. Awais Rathore
Lab No.	11
Tool Used	XAMPP (MySQL / MariaDB) with phpMyAdmin

1. Objective

- Perform numeric rounding and bounding using ROUND, FLOOR, CEIL, and MOD.
- Extract and format date components using YEAR, MONTHNAME, DAYNAME, and DATE_FORMAT.
- Calculate elapsed time using DATEDIFF and TIMESTAMPDIFF.
- Project future dates using DATE_ADD, and combine multiple scalar functions into one query.

2. Database Structure

This lab reuses a Customer/Product schema (scalar_lab_b) for guided practice, then applies the same techniques to a separate Employee database (emp_lab) for the assessment.

2.1 scalar_lab_b

Table	Purpose
Customer	Customer details, including JoinDate and DOB
Product	Product details, including Price and LaunchDate

2.2 emp_lab (assessment)

Table	Purpose
Employee	Employee details used for the Q1+ assessment tasks

3. Key Tasks / Concepts Implemented

- Task B1-B4: Discount/tax calculations with ROUND, and bounding values with FLOOR/CEIL.
- Task B5: Using MOD to isolate odd-numbered IDs.
- Task B6-B7: Extracting year/month/day from a date, and formatting dates with DATE_FORMAT.
- Task B8-B9: Calculating age with TIMESTAMPDIFF, and membership duration with DATEDIFF.
- Task B13-B15: Product market age, projecting a milestone date with DATE_ADD, and combining several scalar functions into one summary string.
- Assessment Q1-Q5: applying string and numeric functions (name cleanup, email generation, masked phone, salary increase) on the Employee table.

4. Implementation (phpMyAdmin)

All numeric/date tasks and the Employee assessment were executed using phpMyAdmin under XAMPP.

Server: 127.0.0.1 » Database: scalar_lab_b » Table: Product

Browse

Structure

SQL

Search

Insert

Export

Import

Privileges

Operations

Tracking

M

-- Task B1: Calculating a 15% discount and rounding to two decimal places. **SELECT** ProdName, Price, ROUND(Price * 0.85, 2) AS DiscountedPrice **FROM** Product;

[Edit inline] [Edit] [Create PHP code]

☐ Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

ProdName

Price

DiscountedPrice

☐ Edit Copy Delete Laptop Pro 15	185000.00	157250.00
☐ Edit Copy Delete Wireless Mouse	2500.00	2125.00
☐ Edit Copy Delete USB-C Cable	800.00	680.00
☐ Edit Copy Delete Office Chair	18500.00	15725.00
☐ Edit Copy Delete Standing Desk	45000.50	38250.43
☐ Edit Copy Delete Notebook A4	350.00	297.50
☐ Edit Copy Delete Ballpoint Pen 10pk	450.00	382.50
☐ Edit Copy Delete Coffee Beans 1kg	1899.99	1614.99
☐ Edit Copy Delete Green Tea Box	650.00	552.50
☐ Edit Copy Delete Bluetooth Speaker	7500.00	6375.00

ROUND query

SELECT ProdName, Price, FLOOR(Price / 1000) AS FloorVal, CEIL(Price / 1000) AS CeilVal **FROM** Product;

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all

Number of rows: 25

Filter rows: Search this table

Sort by key: None

Extra options

ProdName

Price

FloorVal

CeilVal

☐ Edit Copy Delete Laptop Pro 15	185000.00	185	185
☐ Edit Copy Delete Wireless Mouse	2500.00	2	3
☐ Edit Copy Delete USB-C Cable	800.00	0	1
☐ Edit Copy Delete Office Chair	18500.00	18	19
☐ Edit Copy Delete Standing Desk	45000.50	45	46
☐ Edit Copy Delete Notebook A4	350.00	0	1
☐ Edit Copy Delete Ballpoint Pen 10pk	450.00	0	1
☐ Edit Copy Delete Coffee Beans 1kg	1899.99	1	2
☐ Edit Copy Delete Green Tea Box	650.00	0	1
☐ Edit Copy Delete Bluetooth Speaker	7500.00	7	8

FLOOR & CEIL query example (From # B3 task)

```
SELECT CustName, DOB, TIMESTAMPDIFF(YEAR, DOB, CURDATE()) AS Age FROM Customer;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: Filter rows: Sort by key:

Extra options

				CustName	DOB	Age
☐				Ali Khan	1995-04-12	31
☐				Sara Iqbal	1998-11-20	27
☐				HAMZA RAZA	1997-08-05	29
☐				Ayesha Noor	1999-02-14	27
☐				bilal ahmed	2000-06-30	26
☐				Fatima Sheikh	1996-10-25	29
☐				Usman Tariq	2001-03-18	25
☐				Maira Javed	1994-12-09	31

```
SELECT CustName, JoinDate, DATEDIFF(CURDATE(), JoinDate) AS DaysSinceJoin FROM Customer;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: Filter rows: Sort by key:

Extra options

				CustName	JoinDate	DaysSinceJoin
☐				Ali Khan	2022-01-15	1718
☐				Sara Iqbal	2022-04-22	1621
☐				HAMZA RAZA	2023-02-10	1327
☐				Ayesha Noor	2023-05-18	1230
☐				bilal ahmed	2023-09-01	1124
☐				Fatima Sheikh	2024-01-12	991
☐				Usman Tariq	2024-06-30	821
☐				Maira Javed	2024-08-25	765

Output of the age/date-difference queries

```
SELECT EmpID, FullName, UPPER(TRIM(FullName)) AS CleanedName FROM Employee;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: Filter rows: Sort by key:

Extra options

				EmpID	FullName	CleanedName
☐		Edit		Copy		Delete
				2001	ahmad raza	AHMAD RAZA
☐		Edit		Copy		Delete
				2002	Sara Imran	SARA IMRAN
☐		Edit		Copy		Delete
				2003	BILAL KHAN	BILAL KHAN
☐		Edit		Copy		Delete
				2004	Fatima Ali	FATIMA ALI
☐		Edit		Copy		Delete
				2005	Hira Yousaf	HIRA YOUSAF
☐		Edit		Copy		Delete
				2006	Zain Abbas	ZAIN ABBAS
☐		Edit		Copy		Delete
				2007	Mehwish Anwar	MEHWISH ANWAR
☐		Edit		Copy		Delete
				2008	Talha Hussain	TALHA HUSSAIN
☐		Edit		Copy		Delete
				2009	Areeba Yasin	AREEBA YASIN
☐		Edit		Copy		Delete
				2010	Hassan Ahmed	HASSAN AHMED

```
SELECT FullName, Salary, ROUND(Salary * 1.125, 2) AS NewSalary FROM Employee;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: Filter rows: Sort by key:

Extra options

					FullName	Salary	NewSalary
☐		Edit		Copy		Delete	
					ahmad raza	120000.50	135000.56
☐		Edit		Copy		Delete	
					Sara Imran	95000.00	106875.00
☐		Edit		Copy		Delete	
					BILAL KHAN	85000.75	95625.84
☐		Edit		Copy		Delete	
					Fatima Ali	110000.00	123750.00
☐		Edit		Copy		Delete	
					Hira Yousaf	70000.00	78750.00
☐		Edit		Copy		Delete	
					Zain Abbas	78000.40	87750.45
☐		Edit		Copy		Delete	
					Mehwish Anwar	125000.00	140625.00
☐		Edit		Copy		Delete	
					Talha Hussain	60000.00	67500.00
☐		Edit		Copy		Delete	
					Areeba Yasin	90000.99	101251.11
☐		Edit		Copy		Delete	
					Hassan Ahmed	65000.00	73125.00

```
SELECT FullName, DATE_FORMAT(HireDate, '%d-%b-%Y') AS FormattedHireDate FROM Employee;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

				FullName	FormattedHireDate
☐				ahmad raza	01-Sep-2018
☐				Sara Imran	15-Mar-2019
☐				BILAL KHAN	20-Jan-2020
☐				Fatima Ali	10-Nov-2017
☐				Hira Yousaf	05-Apr-2021
☐				Zain Abbas	30-Aug-2022
☐				Mehwish Anwar	22-Jul-2016
☐				Talha Hussain	09-Jan-2023
☐				Areeba Yasin	12-Sep-2019
				Hassan Ahmed	18-Feb-2024

3 Employee assessment queries

5. Conclusion

This lab completed the scalar function toolkit by covering numeric and date/time operations, essential for real-world reporting such as age calculation, discounts, and formatted date display.